Welcome to IT 540
Introduction to Data Science
Week 1: Linear Regression

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Important for our Field!

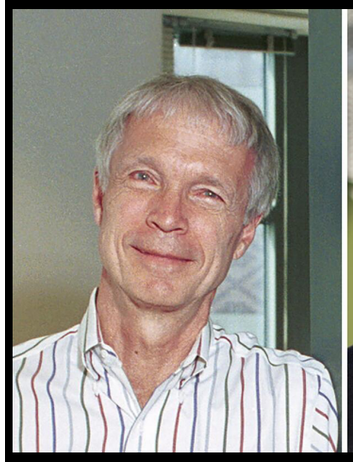
The New York Times

BREAKING NEWS

The Nobel Prize in Physics was awarded to John Hopfield and Geoffrey Hinton for work on machine learning with artificial neural networks.

Tuesday, October 8, 2024 6:04 AM ET

"John Hopfield and Geoffrey Hinton used tools from physics to construct methods that helped lay the foundation for today's powerful machine learning," the Nobel Committee for Physics said. "Machine learning based on artificial neural networks is currently revolutionizing science, engineering and daily life."





Miscellaneous Topics

- We do have a tutor available for this course if you need help, Jordan Schalter. He is VERY good, and has taken this class from me. Please contact Jordan first if you need assistance. His contact info should be in the syllabus.
- Language and Books: Python and books selected by Walsh
- You can ignore references to the R programming language when you see them in the EMC book. We are now using only Python. (However, if you already know R and do not know Python, you may use R. There are assignments for both classes in Moodle.)
- Lecture materials are created by me: Lots of info not in books
- I will also reference my personal research
- I also teach IT 556 Machine Learning, IT 720 Natural Language Processing, and 2 other courses

Most of my past students have NOT had previous programming experience. Even though programming IS a prerequisite for this class, I know some of you have little to no experience.

Some Specific Topics We Will Cover

- Major Machine Learning Topics:
  - Linear and Logistic Regression
  - Classification
  - Clustering
  - Neural Networks
  - Temporal Data Mining
  - Association Rules
  - Natural Language Processing (I will dramatically revise this topic)
- Hadoop Ecosystem and Big Data Computing
- The Art and Practice of Data Science (scattered among the weeks)

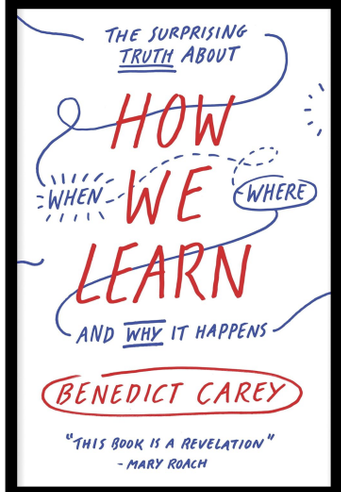
Topics in blue font are NOT covered in our textbooks. Neural networks are THE MOST exciting part of data science and where all the breakthroughs are coming from. Temporal Data Mining will be discussed only briefly in the same week that we discuss Association Rules because those 2 topics are closely related.

I plan to greatly revise the NLP topic before we get to it in Week 9. I will be deleting some material and also adding some that relates to all of the exciting developments in Large Language Models such as ChatGPT and all the others.

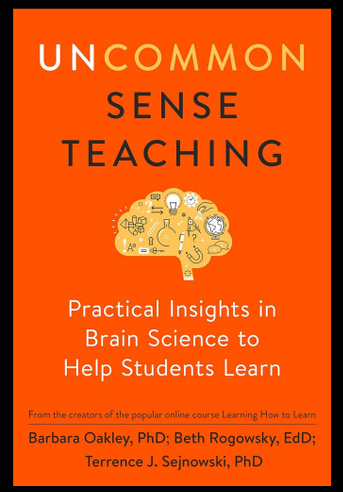
My Objective and Teaching Philosophy:

Objective: Your Success

Teaching Philosophy: Based on Learning Theory



1. **Spaced Repetition**
2. **Varied Retrieval**



That's my only objective for this course. I define your success in terms of (1) learning the material, but even more importantly, (2) getting you EXCITED about data science and machine learning! I really want you to WANT to learn more about this amazing field. To the extent that you do, your life will be enriched and I will gain great satisfaction.

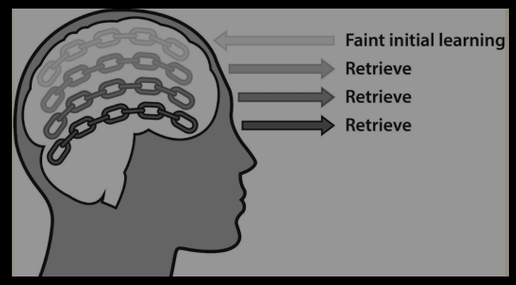
I am highly influenced by research on learning theory. I strongly recommend that you all read:
https://www.amazon.com/How-We-Learn-Brains-Machine-ebook/dp/B07S1K4TB2/ref=sr_1_1
More information in a readable format is this book:
https://www.amazon.com/Uncommon-Sense-Teaching-Practical-Insights/dp/0593329732
My interest began when I took this course several years ago:
https://www.coursera.org/learn/learning-how-to-learn

Spaced Repetition

- Expose the most important ideas multiple times
- Present them in different contexts
- Explain them from different perspectives (e.g. non-technical, technical, simple, complex, etc.)

I did this deliberately when studying for my PhD comprehensive exams long before I knew anything about learning theory.

Varied Retrieval



- The illusion of competence: info is only in short-term memory
- To transfer to long-term memory:
    1. You must <u>actively</u> retrieve memories multiple times
    2. Learn/study in different contexts (e.g. different locations, with/without music, different times, etc.)
    3. Interrupt your own learning, then return to it.
    4. Each act of varied retrieval strengthens the brain's neurons for that memory.

You get the illusion of competence if you memorize something, and repeat it to yourself multiple times. But have you ever had the experience of being interrupted or distracted in the middle of that and then you can no longer remember it? I have had that experience in language learning.
One of the most effective learning techniques is to <u>retrieve</u> the memory from your brain. For example, use flash cards. Or, after you have something in your short-term memory, take a few minutes to write it down without looking. See how much of it you can reproduce accurately, i.e. perform self-testing.
The image is from the book "Uncommon Sense Teaching"

Week 0 Stuff

- Anaconda Python installed? (Or R if you happen to choose that.)
- Did you read/watch Week 0 materials?
- Textbooks purchased?
- My contact info: kgodden@walshcollege.edu, 805-587-1712 (cell)
- Weekly quizzes (my lecture notes, text readings, other)
- ***Special Note***: You can take each quiz TWICE (if you want to) and the best grade of the two is the one that will be recorded.
- 4 assignments
- Final project

Week 1 Advice

- Make sure you go to the “Resources” section of Moodle, which appears before “Week 0”.
- Download items by clicking on them.
- Get them from your downloads folder, and put them somewhere you will remember.
- The pdf file is discussed in my Week 0 video.
- If the .zip file does not automatically expand, then right click on it to create the file called ‘DSBA_code_data’.
    1. Inside, you will find all of the R code and data files used in the EMC textbook.
    2. For EMC Section 3.1 (and all other reading assignments that include R code), you can treat that section as optional, since we no longer use R.

Before we begin…

- Send me an email with your Introductions:
  - Your Name
  - Your occupation
  - Do you have any programming experience? Language(s)?
  - Something interesting about yourself

Traditional Programming vs. Machine Learning

Traditional Programming: Tell the computer what to do



Machine Learning: Show the computer what to do!

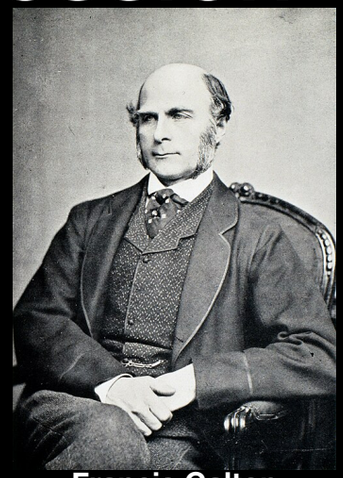


- Peng and Matsui define a model as "something we construct to help us understand the real world" (p.57)
- In more technical terms, a Machine Learning model is a function that makes predictions (either numbers or categories) by mapping input data to labels.
- This function is *learned* by estimating its parameters from the input data and the labels provided during training.

This shows supervised ML, which means we have both the data and its labels. By a 'label', we generally mean either a category (for classification problems) or else a numeric value (for regression problems). The label is what we want to predict for new data that we have not seen before. Your first programming assignment is actually traditional programming because you will implement some equations to compute output values from input data.
You'll learn more about the mechanics of estimating the function's parameters when we discuss neural networks in Week 5. This slide shows how supervised machine learning (ML) works, which is the most common form of ML, where we know the true labels, which help guide or supervise the ML process. We will also study unsupervised ML where we do NOT have the true labels.

Linear Regression

- A linear function is of the form:
  - mx + b, which is equivalent to
  - $\beta_0 + \beta_1 x$

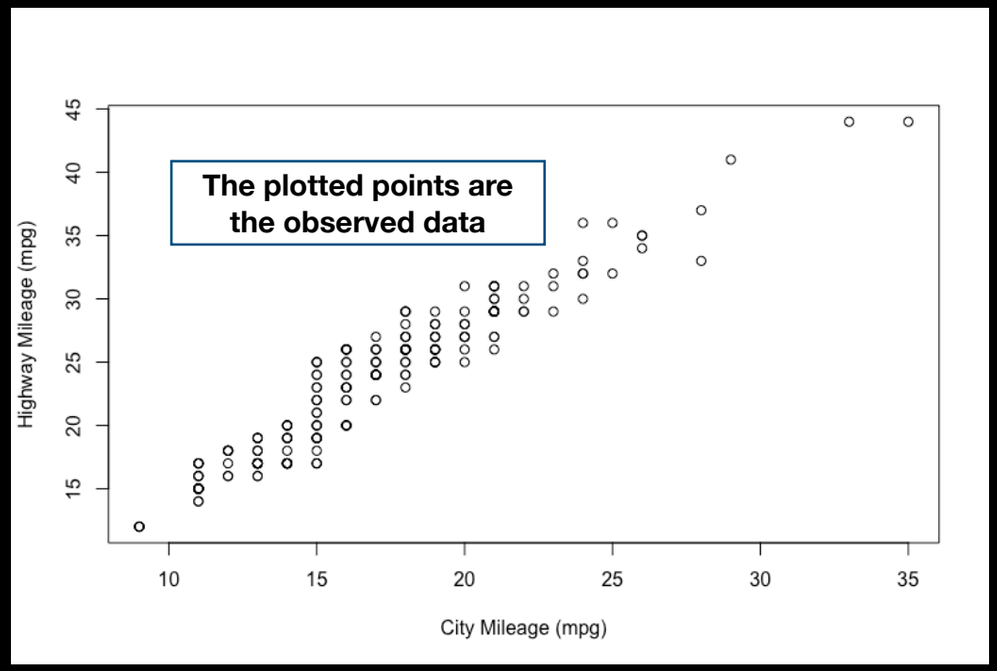


Francis Galton

- From the data points, linear regression learns how to estimate those two parameters β_0 and β_1 in order to fit the *best possible* straight line approximation of the data.
- Regression models are used to predict numbers (as opposed to predicting categories, which we'll discuss next week).

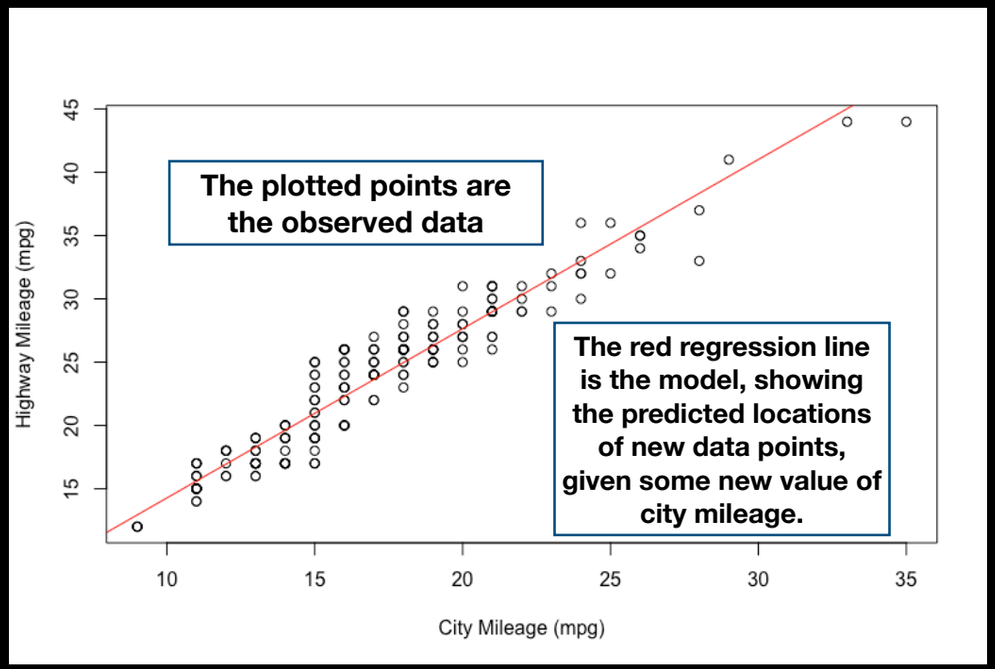
In the 19th century, Francis Galton (image from Wikipedia) was the first to describe and explain the notion of 'regression to the mean'. There are many simple examples, but one is the tendency of tall parents to have children who are shorter than their parents, and short parents tend to have taller children. My own daughters are an example of the former because my wife and I are both tall, and I am an example of the latter because my parents were much shorter than myself.

Linear Regression Example: Vehicle Mileage



It would appear from the data that there is a clear (but not perfect) linear relationship between highway and city mileage of vehicles. The reason for the vertical columns of data on this slide (instead of more randomness) is because the dataset used for this model only measures mileage to the nearest MPG. If it measured to the nearest tenth of a MPG, then you would see the dots more scattered and random looking.

Linear Regression Example: Vehicle Mileage



Method of Least Squares:

- Published just after 1800 by both French mathematician Legendre and German mathematician Gauß (Gauss) on astronomical problems.

- Determine coefficients (parameters) of a linear equation that minimize the mean squared error between the observed data and the predictions.



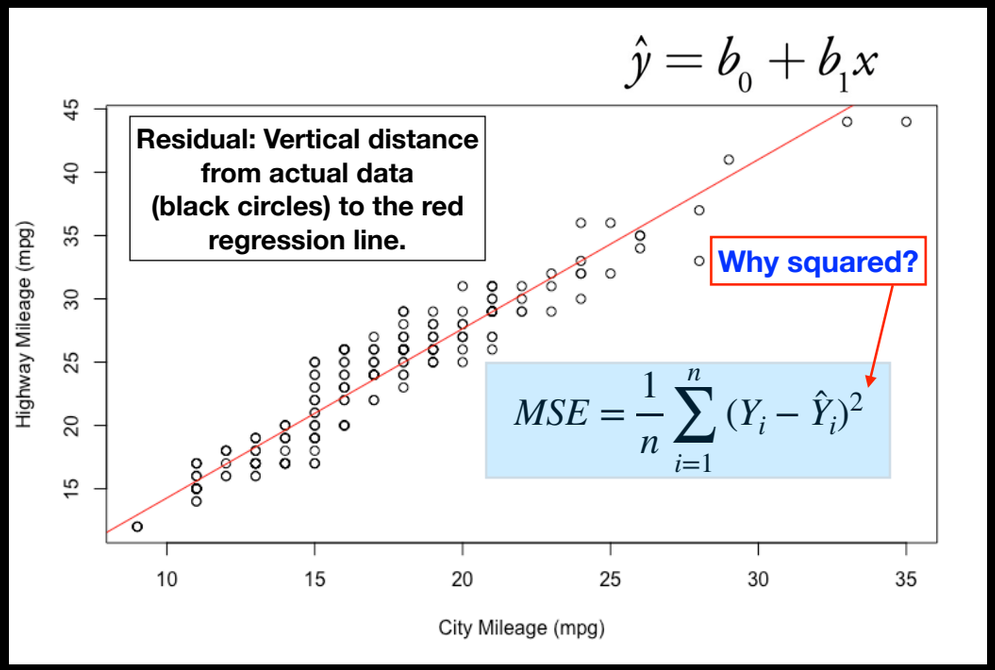
Legendre



Gauß

Linear regression by the method of least squares was introduced by Legendre and Gauss in the 19th century.
The next slide introduces the concept of this method.

Mean Squared Error and Residuals



This is the best possible fit. But what does that mean?

In an earlier slide, I quote Peng and Matsui's definition of a model. This slide says that the model is represented by the red regression line, so try to think how that red line "helps us understand the real world".

How can we give a precise definition of what the *best fit* actually means? By convention the predicted values are shown on the y axis and the predictor variable is shown on the x axis. The value being predicted is called the response (or dependent) variable (in this example, highway mileage) and the information on the x axis is called the predictor or independent variable (in this example, city mileage).

Goal of Machine Learning

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Actual Label → $Y = f(X) + \epsilon$

Predicted Label → $\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 x$

The goal of machine learning is to estimate a target function ‘f’.

In the case of linear regression, this is done by estimating the coefficients β_0 and β_1 in such a way that *minimizes* the MSE.

The observed data Y (the circles in the previous slide) is presumed to be from some function of the input f(X) + epsilon, an error term with mean 0. In the example of the previous slide f( ) is assumed to be linear, but it can be anything in practice.
The estimated points (on the red regression line) are the predicted points obtained by estimating the unknown function f.
Epsilon refers to ‘noise’ or randomness in the actual data values. It is irreducible because no matter how well we estimate function f, epsilon is independent of f. What kinds of things could account for epsilon? (Possible answers: The true function f( ) may not be precisely linear. There may be another unknown independent variable. There may be another variable that is known, but not actually measurable. There could be measurement error of X and Y … etc.)
The general goal of machine learning is to estimate the unknown function f( ) as closely as possible by estimating (in the linear case) the parameters β_0 and β_1 because an accurate estimate of f( ) will help minimize the MSE.

Sidebar ‘Demo’:

- The ML way of estimating parameters

  vs.

- The non-ML way

Watch the recording of the lecture video if you cannot remember what I did here, but I’m showing you how ML generally works by first guessing at parameter values, then improving those guesses vs. the non-ML way which would be to directly program it, e.g. using a ‘closed form’ solution, i.e. an equation that computes the direct answer, which is the subject of Assignment 1.

Note that in the demo I’m using Python, rather than R.

Motivation

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

- Practical goal is to minimize the MSE
- Compare to equation 6-3, p.164 of EMC text (called the Residual Sum of Squares (RSS))
- What terms of the MSE can we control?
- Thus, we seek to minimize $E(Y_i - \hat{Y}_i)^2$, the expected value of the squared residuals.

We may be able to control n (but maybe not), the number of observations we use to estimate f-hat, and in general the greater the value of n, the better. We can also 'control' f-hat itself by the methods we use to estimate it. We cannot control the true underlying function f, nor can we control the error term epsilon. If we just looked at the sum of the residuals instead of the sum of their squares, then it's possible for the sum to be zero even though none of the predicted points are equal to their target values of Y_i By taking the square, the MSE can only equal zero if all predicted points are exactly equal to their targets.
Notice another result of taking the squares of the residuals: Large residuals (predictions that are quite far away from the actual data value) are penalized much more than predictions that are closer to their actual data values. It makes intuitive sense to penalize outliers more than any other residuals.
Note that we have not discussed HOW the machine learning model manipulates the parameter estimates in order to find the smallest possible value for the mean squared error (MSE), but we will discuss that algorithm in Week 5 when we talk about *gradient descent* in the context of neural networks. (Preview: the algorithm is referred to as 'stochastic gradient descent'.)

Reducible
vs.
Irreducible Error

For a given estimate $\hat{f}(X)$ of target function $f(X)$ and a given set X of predictors:

$$
\begin{aligned}
& E(Y_i - \hat{Y}_i)^2 \\
& = E(f(X) + \epsilon - \hat{f}(X))^2 \\
& = (f(X) - \hat{f}(X))^2 + Variance(\epsilon)
\end{aligned}
$$

$(f(X) - \hat{f}(X))^2$ is the *reducible* error, while $Variance(\epsilon)$ is the *irreducible* error.

You can think of epsilon as noise or randomness in the data. (Perhaps due to limitations of our measurement processes, or other causes beyond our control. The variance of epsilon is just the square of the standard deviation of epsilon, and epsilon is assumed to have a mean of 0. This variance is regarded as the ***ir***reducible error — the amount of error that we CANNOT control. The reducible error is just that: by improving our methods, obtaining more data or more precisely measured data, etc. we can reduce the expected squared difference of the observed data and the predicted data.
But, by definition of epsilon, we can never improve the irreducible error, which represents a lower bound on the error.

Training vs. Test Data

- In supervised learning we can divide the available data into 2 sets:
  - Training (generally 70%-80%)
  - Test (the other 20%-30%)
- We create the model using the training data, but really only care about test data MSE.
- Why? (answer is on next slide)

Why? See discussion p. 30 and following in ISLR. Remember that book? It was mentioned in the Week 0 lecture. It's called 'ISLR' because the title is "Intro to Statistical Learning with Applications in R". You can find it here: https://www.statlearning.com/
You will also find a Python version of the book at the same URL.

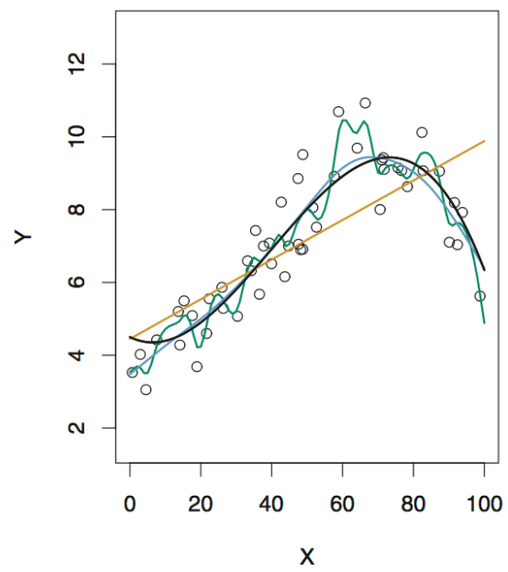
Overfitting the training data leads to the typical training vs. test MSE curves coming up in another slide.
I'm not exactly being truthful in this slide because *in practice*, we will divide the available data into *three* groups, not two. But we will postpone the discussion of this third set, called the validation set, until later in the course.

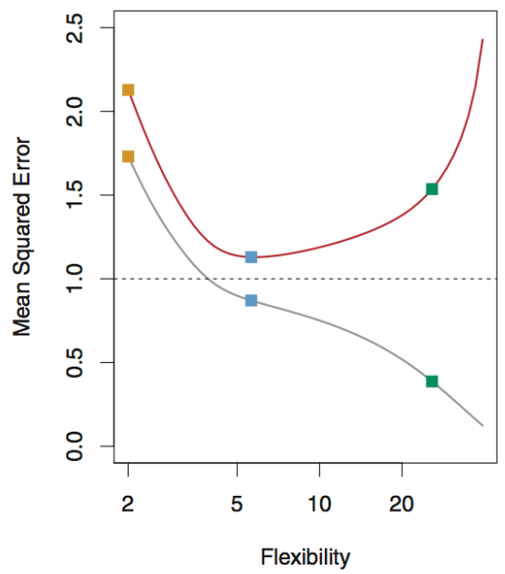
Training Errors vs. Test Errors

- To create a useful model we need to estimate how it will perform in practice.
- The test error is an unbiased estimate of the model's error
- Training error is NOT unbiased. It has an optimistic bias since the final model was selected precisely because that model performed well on the training data.
- This is why we care much more about the test data errors. They are more representative of what we can expect in production.

Reference: p.60 of *Learning from Data: a Short Course,* Abu-Mostafa, et. al. AMLbook.com, 2012

Training Error vs. Test Error*





Flexibility Controls the Tradeoff
between Bias and Variance

Left: The black curve represents the true underlying function and the open circles are the observed data points that we work with to estimate that function. The orange, blue and green curves are models (estimates) of the function with increasing levels of flexibility (decreasing levels of bias). On the right is the MSE by flexibility curve. The gray curve is the MSE of the training data, which always decreases as flexibility increases, and the red curve is the MSE of the test data, which typically has this 'bathtub' shape that decreases and then begins in increase. Why does that happen? To understand, we have to look at the Bias-Variance tradeoff in another slide. And also, what do we mean by flexibility? (Answer as it relates to linear regression: higher-degree polynomials. A more general answer: models with more parameters)

*We can understand bias and variance in terms of human bias and diversity (variance):

How? In people, bias against a person or group of people simply means that the people in question are much more complex than the concept embodied by the bias. The bias does not properly describe the person(s). It's an oversimplification, just like the orange linear regression line in this diagram is an oversimplification of the more complex data points.

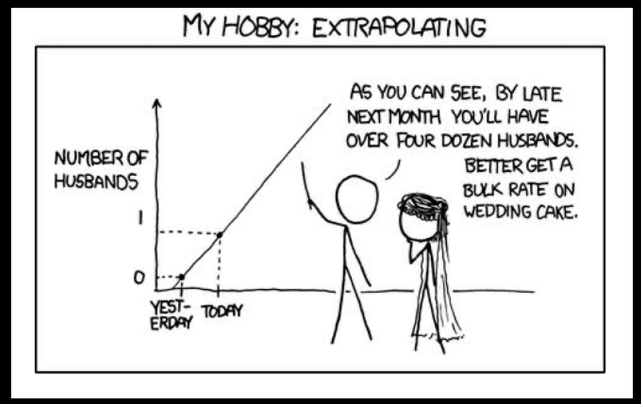
Now think of it in terms of a jury in a criminal trial. You have your own ideas about what evidence is important, but you are also at risk of your own biases. With more people to discuss the evidence, you are adding complexity, flexibility and diversity (variance) of ideas into the discussion. A group is more flexible than a single person and the discussion often changes the minds of most people in one way or another.
As you increase the diversity (variance) of the group, you directly reduce the errors caused by the bias of individuals. This corresponds to the drop in MSE in the top curve on the right when increased flexibility and complexity of the model causes a reduction in the bias.

However, too much flexibility, diversity and variance is also not good. Imagine a jury with 1,000 jurors!! It would be chaotic and there might be factions of people banding together and arguing with each other. That corresponds to the increase of MSE on the right in the same curve.

The best amount of flexibility is, as Buddha advised for life, the middle path between the extremes where there is a balance between bias and variance. That's where the lowest MSE is to be found.

I witnessed this first hand in an attempted murder trial where I was a jury member and saw the quick initial vote of 'guilty' gradually get changed to our final vote of 'innocent' after the deep, lengthy discussion of evidence. It was the added variance of the 12 jury members that led to a unanimous vote that we all agreed on.
The graphic is from ISLR Figure 2.9

Short Break:



What is the primary cause for this linear model's poor reliability?

This cartoon is from xkcd.com, a website that is ultra-famous among nerdy people.

The problem depicted in the cartoon is that a prediction of the future is made without adequate data. (Of course, there is also an absence of common sense!)

Bias-Variance Tradeoff

- There is a proof that the expected *value of the squared difference between* f(x) and $\hat{f}(x)$ for a given input x is the sum of 3 terms:

$$E\left(f(x) - \hat{f}(x)\right)^2 = Variance(\hat{f}(x)) + Bias(\hat{f}(x)) + Variance(\epsilon)$$

- This *expected value of the squared difference means:*

Estimate the target function on each input *x* over *all* possible training sets, taking the overall average across all values of *x.*

This is a very important concept and one that I expect you to become familiar with.
The expected value of the squared difference is the sum of the variance of the estimated function, plus the bias of the estimated function (more precisely, it's the square of the bias), plus the variance of the error term.

Variance of the Estimated Function

$$E\left(f(x) - \hat{f}(x)\right)^2 = \underbrace{Variance(\hat{f}(x))}_{E[\left(\hat{f}_{id}(x) - \bar{f}(x)\right)^2]} + Bias(\hat{f}(x)) + Variance(\epsilon)$$

- The intuition is that each training data set will provide different estimates of the target *f(x).* $\bar{f}(x)$ *is the mean of all those estimates.*
- The variance of the estimated function refers to the expected *squared difference of* all those estimates with their mean.
- If a machine learning model has *high variance*, then the model might change quite a lot if you use training set A vs. training set B.

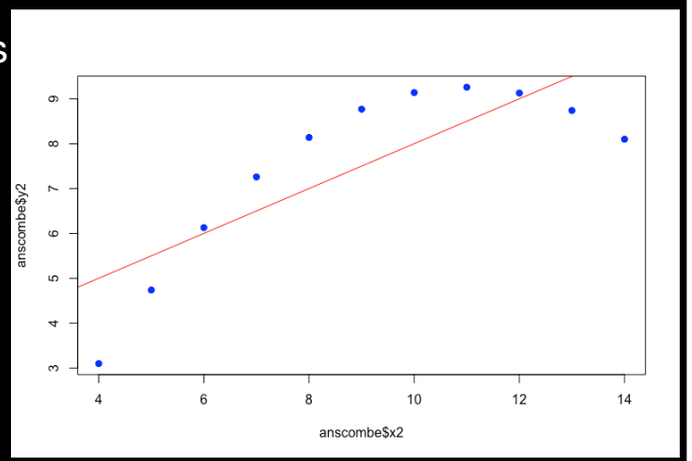
Informally (in non-math terms), variance is the expected amount of change in the estimated function when you change the data that it was estimated from.
Equation adapted from section 2.3.1 Bias and Variance in *Learning from Data: a Short Course,* Abu-Mostafa et. al. AMLbook.com, 2012.
A visual of variance using 5 fingers: Middle finger is the mean of the other 4, if the other 4 are equidistant from that mean. High variance: fingers are spread out, so the squared differences are large. Low variance: fingers are not spread at all, so the squared differences are small.

Bias of Estimated Function

$$E\left(f(x) - \hat{f}(x)\right)^2 = Variance(\hat{f}(x)) + Bias(\hat{f}(x)) + Variance(\epsilon)$$
$$\left(\bar{f}(x) - f(x)\right)^2$$

- Bias is the squared difference of the mean of the function estimates with the actual target function, and is the amount of error created by using a model that is less complex than the actual function f.

- For example, a linear model is not complex enough to describe a quadratic distribution.



Think of the human bias analogy when looking at this diagram. The red regression line is a high bias model — the data is more complex than the model's assumption of a straight line model.

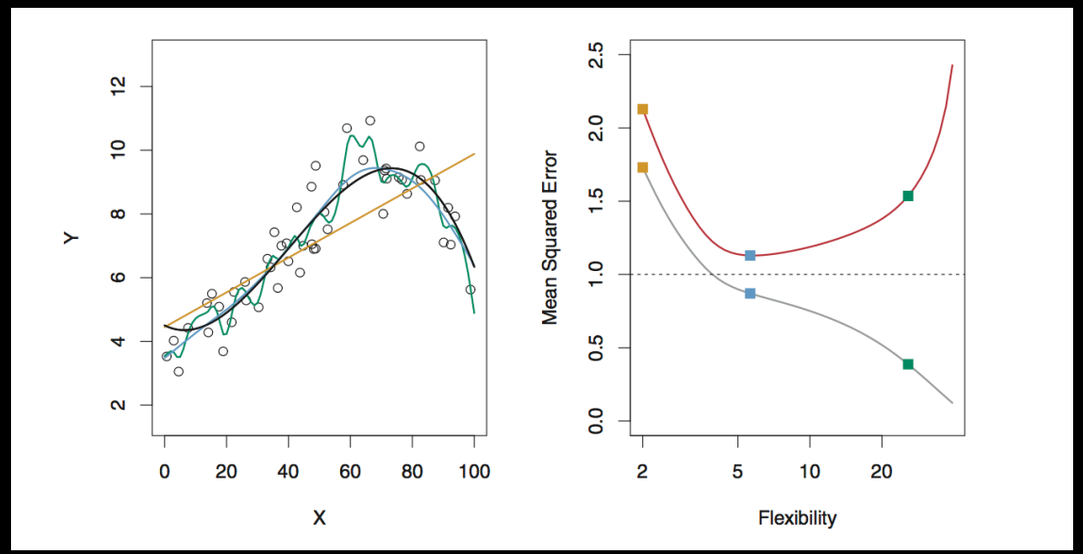
Variance of Error Term

$$E\left(y - \hat{f}(x)\right)^2 = Variance(\hat{f}(x)) + Bias(\hat{f}(x)) + Variance(\epsilon)$$

- The variance of epsilon is the square of its standard deviation, and the mean of epsilon is assumed to be 0.
- Remember that the error term is not anything you can control or influence. It is the 'noise' or randomness of the data values.
- Therefore, we generally do not worry about it since we cannot control it, but it's important to be aware that it exists.

Flexibility of a Model

- A model that is flexible will fit the training data better than a model that is less flexible. Sound good…?



Maybe not, as the graph on the right shows. Same diagram as we discussed several slides back.
Right: The gray curve is the MSE of the training data and red curve is MSE of the test data.

Why is it Called the Bias-Variance Tradeoff?

1. Notice that to obtain a low test MSE, you need to find a model that has *both* low model variance and low model bias.

2. However, as model flexibility (complexity) increases

   - Model variance goes up, but

   - Model bias goes down.

The relationships between model variance and model bias to flexibility are in conflict with one another. This conflict is reflected in the models depicted in the previous slide.
Not on quiz: But with a fixed flexibility, if you add more data to the training set, the variance will decrease, while the bias will remain the same. (This idea is explained on p.67 of the Abu-Mostafa *Learning from Data* book previously referenced.)

Multivariate Linear Regression

- Direct extension of simple linear regression
- As many beta parameters as there are independent variables (plus the β_0 intercept). For *p* predictors:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_p x_p$$

Categorical Variables

- Linear regression requires *numeric* predictors.
- What if you'd like to study the well-known discrepancy of income vs. gender or other inequities?
- Convert the categorical variable to a numeric one by using a dummy variable.

Dummy (Indicator) Variables: Used to Represent Categorical Variables

See EMC pp.166 & 170 for an example of this in R.
For Python, you can use pd.get_dummies in pandas

$x_i = 0$ if the i[th] x value is 'female'

$x_i = 1$ if the i[th] x value is 'male'

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- So if i[th] observation is male the the predicted value of y is $\hat{\beta}_0 + \hat{\beta}_1 x_i = \hat{\beta}_0 + \hat{\beta}_1$
- but if female, then the predicted y is just $\hat{\beta}_0$.

Peng and Matsui, p.15 refers to such variables in an example analysis, showing what happens if you make an error in their interpretation.

Dummy variables are also called 'indicator' variables. If y represents income in the model, then both coefficients β_0 and β_1 would be positive since we would add additional income for men.
However, it does not matter which categorical value is associated with 0 or 1. The interpretation of the beta coefficients will change, however, and the sign and even magnitude of those coefficients may change. E.g. with the dummy var above, β_0 is interpreted as the default (mean) income for women. The value of β_1 is the incremental mean value added to β_0 for men. If we switched the dummy var associations, then β_0 would be the default value for men, and β_1 the incremental value for women, but note that the sign for β_1 would be negative in this case, since women (unfairly) earn less than men, and the magnitude of β_0 would be larger than in the model shown in the slide.

Dummy Variables (2)

- The ‘gender’ variable has 2 values (male, female) and we used 1 binary-valued dummy variable
- If you have more than 2 values for a categorical variable, you need more than 1 dummy variable.
- For a categorical variable with n values, you will need n-1 binary-valued dummy variables.
- The nth value of the categorical variable will be represented by all 0’s, i.e. the intercept $\hat{\beta}_0$ will represent that value.

cf. pp. 85-86, ISLR

Dummy Variables (3)

- For example, if the categorical variable is N. American nationality, then we have 3 values: American, Canadian and Mexican.
- Use 2 binary variables: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}$
  - x_{i1} = 1 for ‘American’ and 0 otherwise.
  - x_{i2} = 1 for ‘Canadian’ and 0 otherwise.
  - ‘Mexican’ is represented by both x_{i1} and x_{i2} being 0, in which case $\hat{\beta}_0$ is the baseline and represents the value of y for Mexicans.
  - It is an error to set both x_{i1} and x_{i2} to 1

Small Technicality:

- The term ‘dummy variable’ is used in statistics, while the term ‘1-hot variable’ is used in data science. They mean almost the same thing.
- But in statistics, you use n-1 dummy variables for n possible categorical values, and when the dummy variables all have the value of 0, then you know that refers to the nth value
- However, in Data Science it is not allowed for all 1-hot variables to have value 0. One of the variables MUST have the value 1.

A few slides back, I told you that pd.get_dummies is used for Python, and that is correct. But it should probably be called ‘pd.get_1hot’ because it creates a 1-hot encoding, not a statistics-based dummy encoding.

Assignment 1:

- First you will create a simple linear model in one predictor.
- You will estimate the parameters of that linear function in two ways:
    1. by solving an equation, i.e. a ‘closed-form’ solution
    2. by using a pre-defined linear regression function
- Compare the parameters estimated by each solution

R's summary( ) Function

```
Call:
lm(formula = mpg ~ disp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max
-4.8922 -2.2022 -0.9631  1.6272  7.2305

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 29.599855   1.229720  24.070  < 2e-16 ***
disp        -0.041215   0.004712  -8.747 9.38e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.251 on 30 degrees of freedom
Multiple R-squared:  0.7183,    Adjusted R-squared:  0.709
F-statistic: 76.51 on 1 and 30 DF,  p-value: 9.38e-10
```

Note: Read the notes to this slide if you will be using R in assignment 1.

This slide shows you the summary of the linear model built from the vehicle data used on a previous slide, but here we are predicting mileage ('mpg') from the size of the engine ('disp' or displacement).

In addition to my own notes below, read EMC pp.168-170 for a discussion of interpreting the output of R's summary( ) function. The t-value is how many std errors the parameter is from 0, and is the 'Estimate' divided by the 'Std. Error'. If X var is not related to Y var, then beta1 would be 0. Here, beta1 ('*disp*') t-val is -8.747, which is a lot. The p-value of the intercept and the beta1 coefficient for disp(lacement) is shown as 'Pr(>|t|). This is the probability that we would observe a value larger than |-8.747| purely by chance, so p-value of 9.38e-10 is VERY unlikely; hence, we can be extremely confident that displacement (the X var) is indeed related to mpg (the Y var) and also that the true value of beta1 is very close to the estimate produced by the regression analysis. A t-statistic (t-value) of about 2.75 corresponds to a p-value of only 0.01 if n=30, and in data science, n is almost always a LOT larger than 30. In this demo with the mtcars dataset, n=32
Notice the significance codes that refer to '***' or '**' or '*' or '.' or ' ' (nothing). A single star by the p-value means that the estimated parameter for that variable (or intercept) is significant at the 0.05 level; two stars is significant at 0.01 level and three stars at the 0.001 level. Generally, you want to see at least '*' to trust the statistical significance of that parameter estimate NOT being due to chance, but '**' is even better and '***' is the best.

The Residual Std Error (RSE) means even if we knew exactly the true param values of the linear equation, our predictions of mpg would still be off by about 3.251 mpg on average. Thus, RSE relates to that epsilon we saw earlier, i.e. the noise. RSE is always in units of the response var, while R^2 gives unit-less characterization, making it easy to compare different models. R^2 ranges from 0-1, where higher values mean more variation of the data is explained by the model. $R^2 = 1 - RSS/TSS$, where RSS is eq.6-3 in EMC and TSS (total sum of squares) is similar but has the mean of Y, rather than the predicted points.

Adjusted R^2 is always slightly smaller than R^2 and the equation is similar but with RSS divided by (n - d - 1), where d is the number of variables actually in the model (as opposed to all the possible variables) and TSS is divided by (n - 1). If you study the math, you'll see that subtracting d is equivalent to a penalty, which lowers R^2 as d

increases. Thus, you get the most useful adj. R^2 when you have a model that uses all and only the significant vars, and you do NOT use the remaining vars, which would just cause you to fit noise in the training set.

The F-stat tells you if ANY of the predictor(s) are related to the response. If $F \approx 1$, then there is little to no relationship, and if $F > 1$, then at least one predictor is related to Y. The F-stat's p-val accommodates presence of all predictors, and can be used in the normal way to evaluate claim that there is a relationship or not, i.e. if 0.05 or less then there is most likely a relationship that is not due to chance. If you include some relatively useless variables, your R^2 values will increase but your F-stat will drop due to their presence.
See chapter 3 of ISLR for more on most of this, and ISLR ch. 6 for explanation of adjusted R^2.

Multivariate Linear Regression

- Direct extension of simple linear regression
- As many beta parameters as there are independent variables (plus the β_0 intercept). For *p* predictors:

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_p x_p$$

- R's lm(y ~ ., data = foo) will create estimates for every coefficient (β_0, β_1, etc.) needed in dataset foo. Also, read the notes for this slide if you use R for assignment 1.

If you are using R for assignment 1, you should also pay attention to this slide because you'll need to have a *formula* similar to '*y ~ .', which* means that *y* is a function of all of the input variables. I have had students in the past who forgot about this slide for assignment 1 and therefore did NOT use 'y~.' in assignment 1 and missed points because of it! So please remember this when you do the assignment. If your input has several predictor variables but you only want to compute the linear regression of y based on ONE of those predictors then you would write $y \sim x_1$, where x_1 is the name of the one variable that is the predictor of *y.* For 2 (but not all) independent predictor variables you would write $y \sim x_1 + x_2$

Linear Regression is Good, but not good is…



If you ever do any video editing, adding text based on the audio that you hear, make sure you listen *carefully*!

This amusing error was created by a video editor who is EXTREMELY good at his job, but did not have any knowledge of data science. So when he heard the individual in this video (a former colleague of mine at Ford) speak the words ‘Linear Regression’ the editor thought he said ‘Linear Aggression’.